Quiz 04. Relapse, vol. 1
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Instructions. In the following problems, the character d is reserved for derivatives and differentials so that $\mathrm{d}\alpha$ is "the differential of α". As is customary, e is the base of the natural logarithm, and $\exp(abc) = e^{abc}$.

Scenario 1-A. Given $y(1) = 2$ and

$$(y^2 - x^2)\,\mathrm{d}x - 2xy\,\mathrm{d}y = 0$$

determine:

1. $y^2(0.25)$;
2. $y^2(0.50)$;
3. $y^2(0.75)$;
4. $y^2(1.25)$;
5. $y^2(1.50)$;
6. $y^2(1.75)$; and
7. $y^2(2.00)$.

Scenario 1-B. Given $y(1) = 1$ and

$$(x^2 + y^2)\,\mathrm{d}x + xy\,\mathrm{d}y = 0$$

determine:

1. $y^2(0.25)$;
2. $y^2(0.50)$;
3. $y^2(0.75)$;
4. $y^2(1.25)$;
5. $y^2(1.50)$;
6. $y^2(1.75)$; and
7. $y^2(2.00)$.

Scenario 2-A. Given $y(1) = 0.25\pi$ and

$$[y^2 - (1 + x^2)y]\,\mathrm{d}x + [x(1 + x^2)]\,\mathrm{d}y = 0$$

for positive x and y, determine:

1. $y\left(\frac{\pi}{6}\right)$;
2. $y\left(\frac{\pi}{3}\right)$; and
3. $y\left(\frac{\pi}{2}\right)$.

Scenario 2-B. Given $y(1) = 0.25\pi$ and

$$[1 - (1 + x^2)y^2]\,\mathrm{d}x - [2x(1 + x^2)y]\,\mathrm{d}y = 0$$

for positive x and y, determine:

1. $y\left(\frac{\pi}{6}\right)$;
2. $y\left(\frac{\pi}{3}\right)$; and
3. $y\left(\frac{\pi}{2}\right)$.

Scenario 3-A. A metal spoon, initially at 25 ºC is placed into a cup of coffee maintained at 85 ºC. After 2 minutes, the temperature of the spoon is measured to be 50 ºC. Under Newtonian heating, estimate the spoon's temperature after:

1. 1.5 minutes;
2. 2.5 minutes;
3. 3.5 minutes;
4. 4.5 minutes; and
5. 5.5 minutes.

Scenario 3-B. A cup of coffee is initially at 90 ºC and is placed in an air-conditioned room maintained at 20 ºC. After 10 minutes, the temperature of the coffee is 60 ºC. Under Newtonian cooling, estimate the coffee temperature after:

1. 1.5 minutes;
2. 2.5 minutes;
3. 3.5 minutes;
4. 4.5 minutes; and
5. 5.5 minutes.

Scenario 4-A. Chemical waste from the electrolysis process of a copper refinery plant is stored in a 1000-gal-capacity, well-mixed tank for dilution and discharge. The tank initially contains 512 gal of solution with 200 oz of electrolytes. Fresh water is fed to the tank at a rate of 4 gpm, while the discharge flows at 1 gpm. Determine:

1. the amount (in oz) of electrolytes in the tank after a quarter of an hour;
2. the amount (in oz) of electrolytes in the tank after half an hour;
3. the amount (in oz) of electrolytes in the tank after an hour;
4. the time (in min) it takes for the tank to be at the brink of overflowing; and
5. the amount (in oz) of electrolytes in the tank as it starts to overflow.

Scenario 4-B. Industrial waste from a copper refinery plant is stored in a 500-L-capacity, well-mixed tank for washing and discharge. The tank initially contains 400 L of solution with 200 kg of electrolytes. Fresh water is fed to the tank at a rate of 2 L per minute, while the discharge flows at 6 L per minute. Determine:

1. the amount (in kg) of electrolytes in the tank after a quarter of an hour;
2. the amount (in kg) of electrolytes in the tank after half an hour;
3. the amount (in kg) of electrolytes in the tank after an hour;
4. the time (in min) it takes for the tank to be empty; and
5. the amount (in kg) of electrolytes in the tank ten minutes before it empties out.

Scenario 5-A. A parallel RC circuit comprises a 1-ohm resistor and a 0.5-farad capacitor. The circuit is driven by a time-varying source supplying $16e^{-4t}$ amperes. The initial capacitor voltage is -5 volts. Determine the magnitude of the capacitor voltage after:

1. 100 ms;
2. 250 ms;
3. 500 ms;
4. 750 ms; and
5. 1000 ms.

Scenario 5-B. A parallel RC circuit comprises a 50-ohm resistor and a 0.01-farad capacitor. The circuit is driven by a time-varying source supplying $5e^{-2t}$ amperes. The initial capacitor voltage is -10 volts. Determine the magnitude of the capacitor voltage after:

1. 100 ms;
2. 250 ms;
3. 500 ms;
4. 750 ms; and
5. 1000 ms.

Marking. Each numbered item above is worth two (2) points. The total percentage rating a student gets for this Activity is computed as $0.59(0.9^m r) + 0.41p$, where m is the number of unmet requirements and violated instructions, r is the raw score (in percent), and p is the peer rating (in percent) by his/her groupmates.

Submission. Answers shall be communicated through the accompanying Google Form. Peer ratings shall be communicated via the submission platform in the Google Classroom.